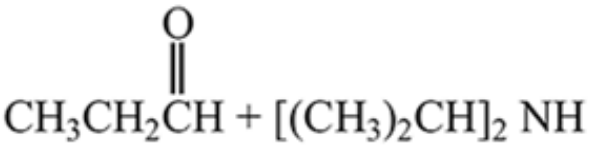
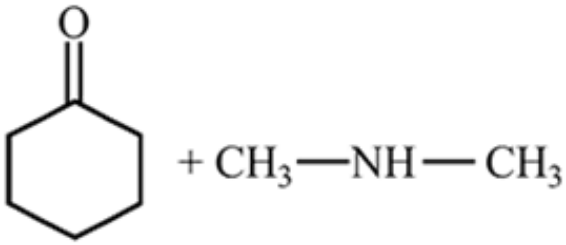
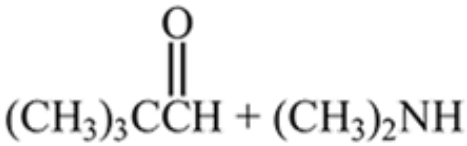
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following pairs of reactants is most effective in forming an enamine?

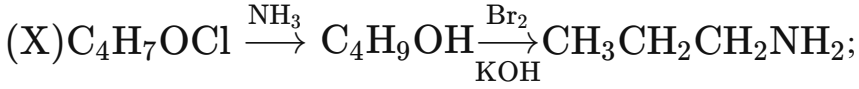
(A) 

(B) 

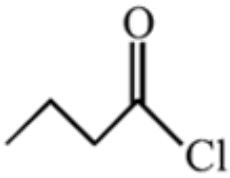
(C) 

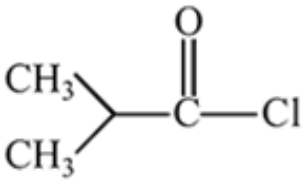
(D) None of these form an enamine.

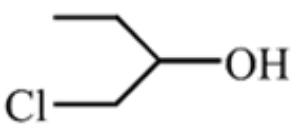
Q.2

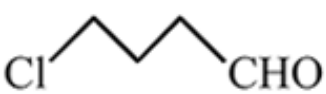


compound (X) is:

(A) 

(B) 

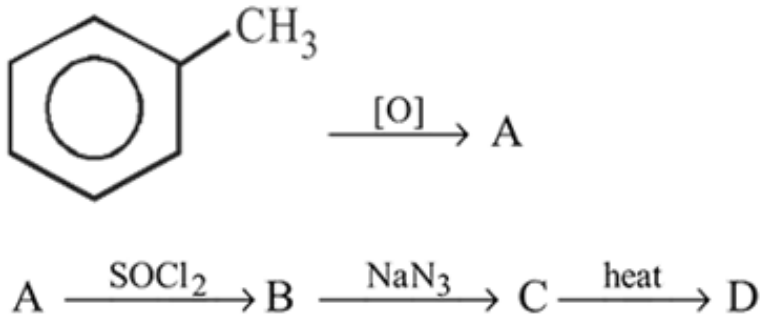
(C) 

(D) 

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Q.3

In the following sequence of reactions, what is D?



(A) Primary amine

(B) An amide

(C) Phenyl isocyanate

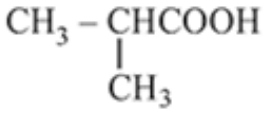
(D) A higher hydrocarbon

Q.4

An organic compound 'A'on treatment with NH_3 gives ' B ' which on heating gives ' C', $^{\circ}C$ ' when treated with Br_2 in the presence of KOH produces ethylamine. Compound 'A' is:

(A) CH_3COOH

(B) $CH_3CH_2CH_2COOH$

(C) 

(D) CH_3CH_2COOH

Q.5

An organic amino compound reacts with aqueous nitrous acid at low temperature to produce an oily nitrosoamine. The compound is

(A) CH_3NH_2

(B) $CH_3CH_2NH_2$

(C) $CH_3CH_2NHCH_2CH_3$

(D) $(CH_3CH_2)_3\ N$

Q.6

Which of the following statements are correct ?

(i) In Sandmeyer reaction nucleophiles like Cl^-, Br^- and CN^- are indroduced in benzene ring in the presence of Cu^+ ion

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(ii) In Gattermann reaction nucleophiles are introduced in benzene ring in the presence of copper powder and HCl.

(iii) The yield in Gattermann reaction is found to be better than Sandmayer reaction.

(A) (i) and (ii)

(B) (i), (ii) and (iii)

(C) (ii) and (iii)

(D) (i) and (iii)

Q.7

Consider the following sequence of reactions:

Compound $[A] \xrightarrow{\text{Reduction}} [B] \xrightarrow{HNO_2} CH_3CH_2OH$

The compound $[A]$ is

(A) CH_3CH_2CN

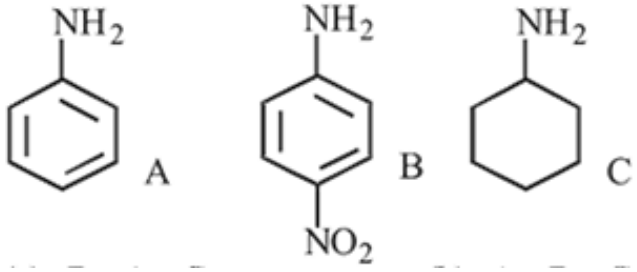
(B) CH_3NO_2

(C) CH_3NC

(D) CH_3CN

Q.8

The correct order of basicity of the following compounds



(A) $B > A > C$

(B) $A > B > C$

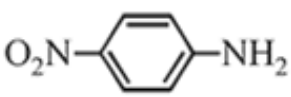
(C) $C > A > B$

(D) $C > B > A$

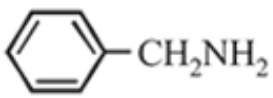
Q.9

Which of the following compounds is most basic?

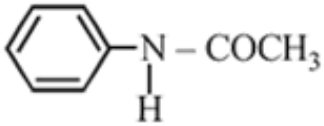
(A)

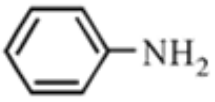


(B)



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(C) 

(D) 

Q.10

Match the compounds given in List - I with their characteristic reactions given in List - II. Select the correct option.

List - I (Compounds)	List - II (Reactions)
(1) $CH_3CH_2CH_2CH_2NH_2$	(i) alkaline hydrolysis
(2) $CH_3C \equiv CH$	(ii) with KOH (alcohol) and $CHCl_3$ produces bad smell
(3) $CH_3CH_2COOCH_3$	(iii) gives white ppt. with ammonical $AgNO_3$
(4) $CH_3CH(OH)CH_3$	(iv) with Lucas reagent cloudiness appears after 5 minutes

(A) 1-(iv) 2-(ii) 3-(iii) 4-(i)

(B) 1-(ii) 2-(i) 3-(iv) 4-(iii)

(C) 1-(iii) 2-(ii) 3-(i) 4-(iv)

(D) 1-(ii) 2-(iii) 3-(i) 4-(iv)

Q.11

Which one of the following on reduction with lithium aluminium hydride yields a secondary amine?

(A) Methyl isocyanide

(B) Acetamide

(C) Methyl cyanide

(D) Nitroethane

Q.12

Which reaction sequence would be best to prepare 3-chloroanilne from benzene?

(A) Chlorination, nitration, reduction

(B) Nitration, chlorination, reduction

(C) Nitration, reduction, chlorination

(D) Nitration, reduction, acetylation, chlorination, hydrolysis

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Q.13

The reduction of nitro compounds is most preferred in the presence of

(A) Pd/H_2 in ethanol

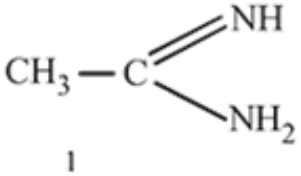
(B) $Sn + HCl$

(C) finely divided Ni

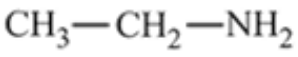
(D) iron scrap and HCl

Q.14

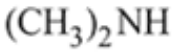
The correct order of basicities of the following compounds is:



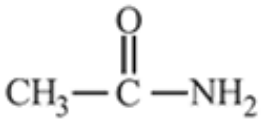
1



2



3



4

(A) $2 > 1 > 3 > 4$

(B) $1 > 3 > 2 > 4$

(C) $3 > 1 > 2 > 4$

(D) $1 > 2 > 3 > 4$

Q.15

Which of the following reactions will not give a primary amine?

(A) $CH_3CONH_2 \xrightarrow{Br_2/KOH}$

(B) $CH_3CN \xrightarrow{LiAlH_4}$

(C) $CH_3NC \xrightarrow{LiAlH_4}$

(D) $CH_3CONH_2 \xrightarrow{LiAlH_4}$

Q.16

Indicate which nitrogen compound amongst the following would undergo Hoffmann reaction (i.e.., reaction with Br and strong KOH) to furnish the primary amine $(R - NH_2)$

(A) $RCONHCH_3$

(B) $RCOONH_4$

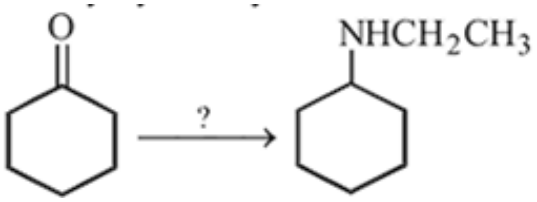
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(C) $RCONH_2$

(D) $R - CO - NHOH$

Q.17

Reagents capable of converting cyclohexanone to N-ethyl cyclohexylamine is



(A) CH_3CH_2Br and NH_3

(B) $CH_3CH_2NH_2$ and H_2/Pt

(C) $CH_3CH = O$ and NH_3

(D) $LiAlH_4$ followed by H_2O and then CH_3CH_2Br

Q.18

Which of the following gives propylamine upon hydrolysis?

(A) $CH_3CH_2C \equiv N$

(B) $(CH_3CH_2CH_2)_2\,NH$

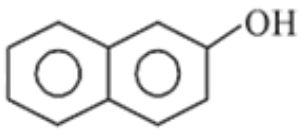
(C)



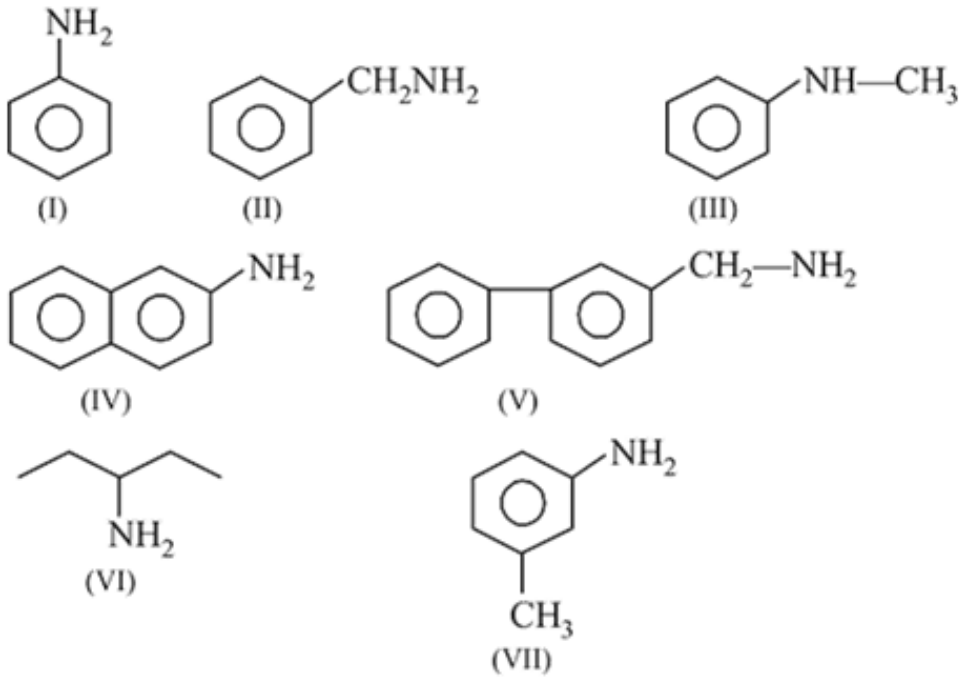
(D) $CH_3CH_2 - CH = NH$

Q.19

Number of amines which will form colourful dye with HNO_2
and



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(A) 2

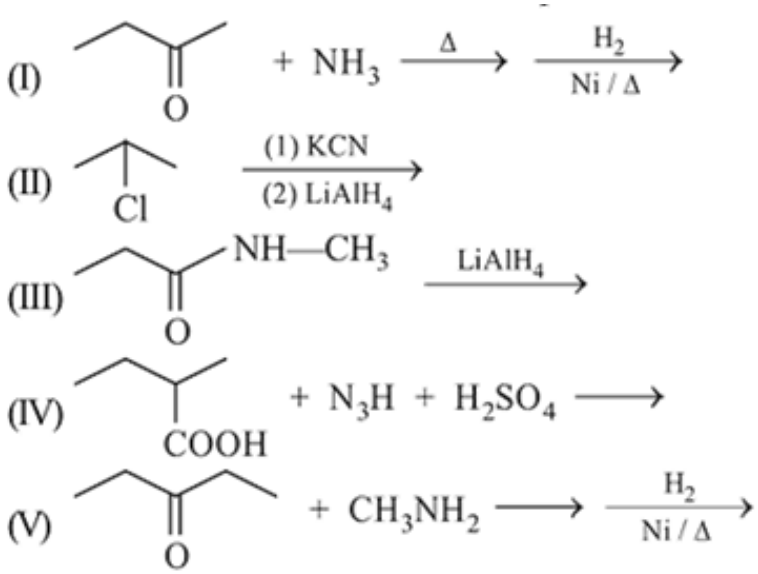
(B) 4

(C) 6

(D) 3

Q.20

Total Number of reactions in which product is 1° amine



(A) 3

(B) 4

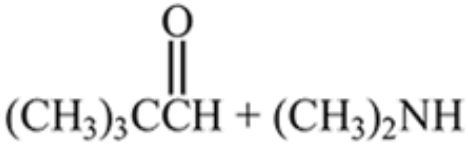
(C) 1

(D) 2

Answers & Solutions

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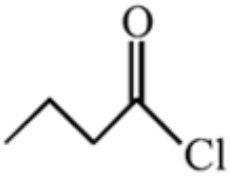
Q.1 Answer:



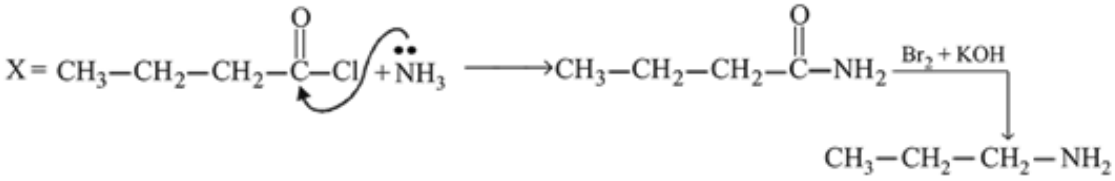
Solution:

2° amine must be less sterically hindered. In option (b) and (c) 2° amine is being used but due to more reactivity of aldehyde compared to cyclohexanone. Option (c) is correct.

Q.2 Answer:



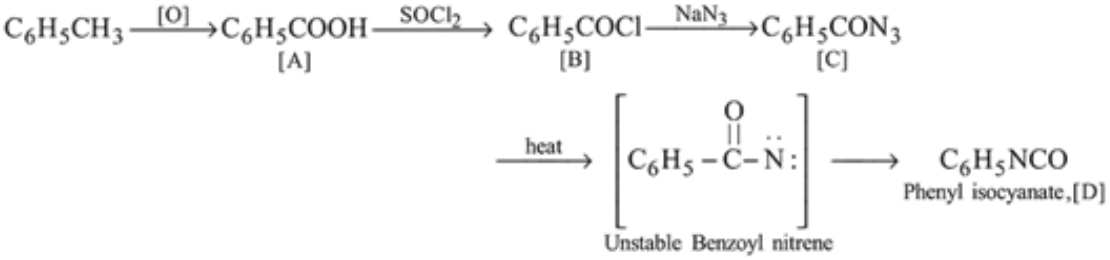
Solution:



Q.3 Answer:

Phenyl isocyanate

Solution:



Q.4 Answer:

CH_3CH_2COOH

Solution:

Since, C when heated with Br_2 in presence of KOH produces ethylamine, hence it must be propanamide and hence the organic compound (A) will be propanoic acid. The reactions follows.

$CH_3 - CH_2 - COOH \xrightarrow{NH_3}$

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(A)

$$\underset{\text{(B)}}{CH_3-CH_2-COO^-NH_4^+} \xrightarrow{\Delta}$$

$$\underset{\text{(C)}}{CH_3-CH_2-CONH_2} \xrightarrow[\substack{\text{Hoffmann}\\\text{bromamide}\\\text{reaction}}]{KOH+Br_2}$$

$$\underset{\text{(Ethylamine)}}{CH_3-CH_2-NH_2}$$

Q.5 Answer:

$CH_3CH_2NHCH_2CH_3$

Solution:

Since the organic amino compound on reaction with nitrous acid at low temperature produces an oily nitrosoamine so the organic amino compound is a secondary aliphatic amines.

Q.6 Answer:

(i) and (ii)

Solution:

The yield in Sandmayer reaction is found to be better than Gattermann reaction.

Q.7 Answer:

CH_3CN

Solution:

$$CH_3CN \xrightarrow{\text{Reduction}} CH_3-CH_2-NH_2$$

$$CH_3-CH_2-OH + N_2 + H_2O \leftarrow$$

Q.8 Answer:

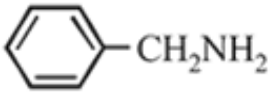
$C > A > B$

Solution:

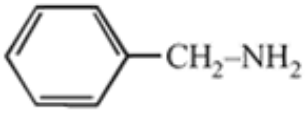
Aliphatic amines are more basic than aromatic amines. Resonance decreases the basic character due to delocalisation of shared pair of electrons on nitrogen within benzene nucleus.

Q.9 Answer:

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Solution:



compound is most basic due to localized lone pair of electron on nitrogen atom while other compounds have delocalized lone pair of electron.

Q.10 Answer:

1-(ii) 2-(iii) 3-(i) 4-(iv)

Solution:

(1) $CH_3CH_2 - CH_2 - CH_2 - NH_2$

(ii) with KOH **(alcohol) and** $CHCl_3$ **produces bad smell**

(2) $CH_3C \equiv CH$

(iii) gives white ppt with ammonical $AgNO_3$

(3) $CH_3CH_2COOCH_3$

(i) alkaline hydrolysis

(4) $CH_3CHOH - CH_3$

(iv) with Lucas reagent cloudiness appears after 5 minutes

Q.11 Answer:

Methyl isocyanide

Solution:

Reduction of alkyl isocyanides in presence of $LiAlH_4$ **yields secondary amines containing methyl as one of the alkyl group.**

$$R - N \equiv C + 4[H] \xrightarrow{LiAlH_4} \underset{2^\circ \text{amine}}{R - NH - CH_3}$$

e.g., $CH_3 - N \geqslant C + 4[H] \xrightarrow{LiAlH_4} \underset{\text{dimethyl amine}}{CH_3 - NH - CH_3}$

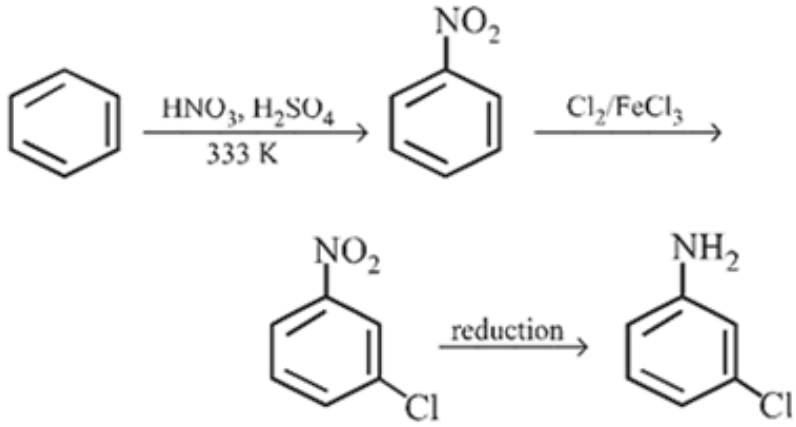
whereas, alkyl cyanides give 1° **amine on reduction.**

Q.12 Answer:

Nitration, chlorination, reduction

Solution:

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(a) When chlorination is done earlier than nitration, chlorobenzene formed at first step would introduce $-NO_2$ group in ortho-position, not in m -position
(b) Again if $-NO_2$ group is reduced earlier than the chlorination step, $-NH_2$ group formed on reduction will again introduce $-Cl$ in o-position

Q.13 Answer:

iron scrap and HCl

Solution:

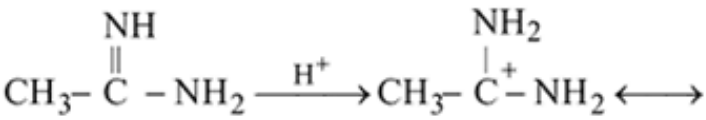
Reduction with iron scrap and hydrochloric acid is preferred because $FeCl_2$ formed gets hydrolysed to release hydrochloric acid during the reaction. Thus, only a small amount of hydrochloric acid is required to initiate the reaction.

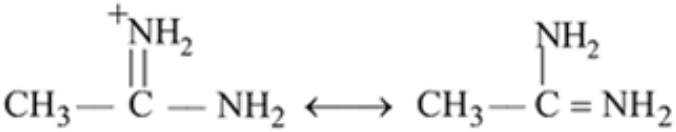
Q.14 Answer:

$1 > 3 > 2 > 4$

Solution:

Guanidine is most basic because its conjugate acid is stabilized by two equivalent resonance structures,

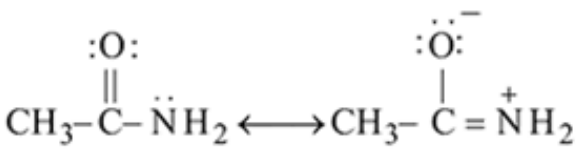




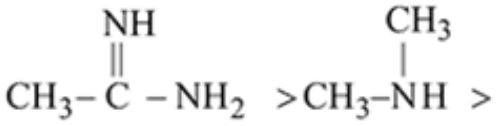
Further 2° amines are more basic than 1°, while amides

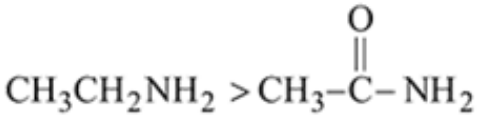
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are least basic due to delocalisation of the lone pair of electrons on N



Thus the correct order of basicities is





Q.15 Answer:

$CH_3NC \xrightarrow{LiAlH_4}$

Solution:

CH_3NC **(methyl isocyanide) on reduction with** $LiAlH_4$ **gives secondary amine.**

Q.16 Answer:

$RCONH_2$

Solution:

Only 1° **amides (i.e.** $RCONH_2$ **) in the present case undergo Hofmann bromamide reaction.**

$RCONH_2 + Br_2 + 4KOH \longrightarrow$

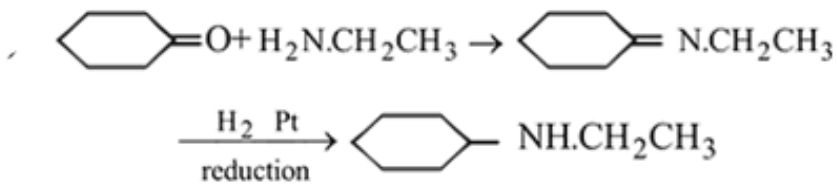
$R - NH_2 + 2KBr + K_2CO_3 + 2H_2O$

(Hofmann's bromamide reaction)

Q.17 Answer:

$CH_3CH_2NH_2$ and H_2/Pt

Solution:



Q.18 Answer:

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Solution:

Amides are hydrolysed to give acids and NH_3 or amines.

Q.19 Answer:

3

Solution:

Amines I, IV, VII will form azo dye.

Q.20 Answer:

3

Solution:

I, II and IV.

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